Exercise Set 10.3 (Page 477)

1. Evaluate in SIX different ways

$$I = \iiint_S (x + 2y + 4z)\, dx\, dy\, dz, \text{ where } S \text{ is defined by}$$

$$1 \le x \le 2, -1 \le y \le 0, 0 \le z \le 3$$

Sol. $$I = \int_1^2 \int_{-1}^0 \int_0^3 (x + 2y + 4z)\, dz\, dy\, dx$$

$$= \int_1^2 \int_{-1}^0 [(x + 2y)\, z + 2z^2]_0^3\, dy\, dx$$

$$= \int_1^2 \int_{-1}^0 (18 + 3x + 6y)\, dy\, dx = \int_1^2 [18y + 3xy + 3y^2]_{-1}^0\, dx$$

$$= \int_1^2 (15 + 3x)\, dx = \left[15x + \frac{3x^2}{2}\right]_1^2 = \frac{39}{2}.$$

The other five orders of integration are similar.

2. Evaluate $I = \int_0^4 \int_0^{4-x} \int_0^{4-x-y} dz\, dy\, dx$. Also change the order of integration so that z-integration is performed last and find its value.

Sol. $I = \int_0^4 \int_0^{4-x} (4 - x - y)\, dy\, dx = \int_0^4 \left[4y - xy - \frac{y^2}{2}\right]_0^{4-x} dx$

$$= \int_0^4 \left[4(4-x) - x(4-x) - \frac{(4-x)^2}{2}\right] dx$$

$$= \int_0^4 \left(8 - 4x + \frac{x^2}{2}\right) dx = \left[8x - 2x^2 + \frac{x^3}{6}\right]_0^4 = \frac{32}{3}.$$

If z – integration is performed last, then

$$I = \int_0^4 \int_0^{4-z} \int_0^{4-y-z} dx\, dy\, dz \quad \text{and} \quad I = \int_0^4 \int_0^{4-z} \int_0^{4-x-z} dy\, dx\, dz.$$

$$\int_0^4 \int_0^{4-z} \int_0^{4-y-z} dx\, dy\, dz = \int_0^4 \int_0^{4-z} (4 - y - z)\, dy\, dz$$

$$= \int_0^4 4y - \frac{y^2}{2} - yz \Big|_0^{4-z} dz = \int_0^4 \left[4(4-z) - \frac{1}{2}(4-z)^2 - z(4-z)\right] dz$$

$$= \int_0^4 \left(8 - 4z + \frac{z^3}{2}\right) dz = 8z - 4\frac{z^2}{2} + \frac{z^3}{6}\Big]_0^4 = 32 - 32 + \frac{64}{6} = \frac{32}{3}$$

Similarly,

$$\int_0^4 \int_0^{4-z} \int_0^{4-x-z} dy\, dx\, dz = \frac{32}{3}.$$

Evaluate (Problems 3 – 10):

3. $\int_0^2 \int_0^1 \int_0^1 xyz\sqrt{2 - x^2 - y^2}\, dx\, dy\, dz$

Sol. $I = \int_0^2 \int_0^1 yz\left[-\frac{1}{3}(2 - x^2 - y^2)^{3/2}\right]_0^1 dy\, dz$

$$= \int_0^2 \int_0^1 yz\left[-\frac{1}{3}(1 - y^2)^{3/2} + \frac{1}{3}(2 - y^2)^{3/2}\right] dy\, dz$$

$$= \int_0^2 z\left[\frac{1}{15}(1-y^2)^{5/2} - \frac{1}{15}(2-y^2)^{5/2}\right]_0^1 dz$$

$$= \int_0^2 z\left(\frac{4\sqrt{2}}{15} - \frac{2}{15}\right) dz = \left[\left(\frac{4\sqrt{2}}{15} - \frac{2}{15}\right)\frac{z^2}{2}\right]_0^2$$

$$= 2\left(\frac{4\sqrt{2}}{15} - \frac{2}{15}\right) = \frac{4}{15}(2\sqrt{2} - 1).$$

4. $\displaystyle\int_0^a \int_0^{\sqrt{a^2-y^2}} \int_0^{\sqrt{a^2-x^2-y^2}} x \, dz\, dx\, dy$

Sol. $\displaystyle I = \int_0^a \int_0^{\sqrt{a^2-y^2}} x\,[z]_0^{\sqrt{a^2-x^2-y^2}} dx\, dy$

$$= \int_0^a \int_0^{\sqrt{a^2-y^2}} x\sqrt{a^2-x^2-y^2}\, dx\, dy$$

$$= \int_0^a \left[-\frac{1}{3}(a^2-x^2-y^2)^{3/2}\right]_0^{\sqrt{a2-y2}} dy$$

$$= \int_0^a -\frac{1}{3}[0-(a^2-y^2)^{3/2}]\, dy = \frac{1}{3}\int_0^a (a^2-y^2)^{3/2} dy$$

Put $y = a\sin\theta$ so that $dy = a\cos\theta\, d\theta$ and

$$I = \frac{1}{3}\int_0^{\pi/2} a^4\cos^4\theta\, d\theta = \frac{a^4}{3}\left[\frac{3}{4.\,2}\cdot\frac{\pi}{2}\right] = \frac{\pi a^4}{16}.$$

5. $\displaystyle\int_0^2 \int_0^{\sqrt{4-z^2}} \int_{y^2+z^2-4}^{4-y^2-z^2} dx\, dy\, dz$

Sol. $\displaystyle I = \int_0^2 \int_0^{\sqrt{4-z^2}} [x]_{y^2+z^2-4}^{4-y^2-z^2} dy\, dz$

$$= 2\int_0^2 \int_0^{\sqrt{4-z^2}} (4 - z^2 - y^2)\, dy\, dz = 2\int_0^2 \left[(4 - z^2)\, y - \frac{y^3}{3}\right]_0^{\sqrt{4-z^2}} dz$$

$$= \frac{4}{3}\int_0^2 (4 - z^2)^{3/2}\, dz$$

Put $z = 2\sin\theta$ so that $dz = 2\cos\theta\, d\theta$ and

$$= I = \frac{4}{3}\int_0^{\pi/2} 8.2 \cos^4\theta\, d\theta = \frac{64}{3} \cdot \frac{3}{4.\ 2} \cdot \frac{\pi}{2}\ 4\pi.$$

6. $\iiint_S z\, dx\, dy\, dz$, S bounded by

$$z = \sqrt{x^2 + y^2},\ z = 0, x = \pm 1, y = \pm 1.$$

Sol. $$I = \int_{-1}^{1}\int_{-1}^{1}\int_{0}^{\sqrt{x^2+y^2}} z\, dz\, dy\, dx = \int_{-1}^{1}\int_{-1}^{1}\left[\frac{z^2}{2}\right]_0^{\sqrt{x^2+y^2}} dy\, dx$$

$$= \frac{1}{2}\int_{-1}^{1}\int_{-1}^{1} (x^2 + y^2)\, dy\, dx = \frac{1}{2}\int_{-1}^{1}\left[x^2 y + \frac{y^3}{3}\right]_{-1}^{1}$$

$$= \frac{1}{2}\int_{-1}^{1}\left(2x^2 + \frac{2}{3}\right) dx = \frac{1}{2}\left[2\frac{x^3}{3} + \frac{2}{3}x\right]_{-1}^{1} = \frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3}.$$

7. $\iiint_S 15x^2 z^2\, dx\, dy\, dz$, S bounded by

$$x^2 + y^2 = 1, x^2 + z^2 = 1.$$

Sol. $$I = \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} 15z^2x^2\, dz\, dy\, dx$$

$$= 15\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \left[x^2 \frac{z^3}{3}\right]_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy\, dx$$

$$= 15\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{3}x^2 (1-x^2)^{3/2}\, dy\, dx$$

$$= 10\int_{-1}^{1} \left[x^2 (1-x)^{3/2} y\right]_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dx$$

$$= 10\int_{-1}^{1} 2x^2(1-x^2)^{3/2}\sqrt{1-x^2}\, dx$$

$$= 20\int_{-1}^{1} x^2 (1-x^2)^2\, dx = 20\int_{-1}^{1} (x^2 - 2x^4 + x^6)\, dx$$

$$= 20\left[\frac{x^3}{3} - \frac{2x^5}{5} + \frac{x^7}{7}\right]_{-1}^{1}$$

$$= 20\left[\frac{1}{3} - \frac{2}{5} + \frac{1}{7} - \left(-\frac{1}{3} + \frac{2}{5} - \frac{1}{7}\right)\right] = 20 \times 2\left[\frac{8}{3\times 5\times 7}\right] = \frac{64}{21}.$$

8. $\iiint_S x^2 y^2 z\, dx\, dy\, dz$, S defined by.

$$0 \le z \le x^2 - y^2,\ 0 \le x \le 1,\ 0 \le y \le 1$$

Sol. $I = \int_0^1\int_0^1\int_0^{x^2-y^2} x^2y^2 z\, dz\, dy\, dx$

$$= \int_0^1\int_0^1 \left[x^2 y^2 \frac{z^2}{2}\right]_0^{x^2-y^2} dy\, dx = \frac{1}{2}\int_0^1\int_0^1 x^2 y^2 (x^4 - 2x^2y^2 + y^4)\, dy\, dx$$

$$= \frac{1}{2}\int_0^1 \left[x^6\frac{y^3}{3} - 2x^4\frac{y^5}{5} + x^2\frac{y^7}{7}\right]_0^1 dx = \frac{1}{2}\int_0^1 \left(\frac{x^6}{3} - \frac{2x^4}{5} + \frac{x^2}{7}\right) dx$$

$$= \frac{1}{2}\left[\frac{x^7}{21} - \frac{2x^5}{25} + \frac{x^3}{21}\right]_0^1 = \frac{1}{2}\left[\frac{1}{21} - \frac{2}{25} + \frac{1}{21}\right] = \frac{4}{525}.$$

9. $\iiint_S (x+1)\, dx\, dy\, dz$, S defined by

$y = 0, y = x$ for $0 \le x \le 1$ and $-y^2 \le z \le x^2$.

Sol. $$I = \int_0^1 \int_0^x \int_{-y^2}^{x^2} (x+1)\, dz\, dy\, dx = \int_0^1 \int_0^x [(x+1)z]_{-y^2}^{x^2}\, dy\, dx$$

$$= \int_0^1 \int_0^x (x+1)(x^2+y^2)\, dy\, dx = \int_0^1 \left[(x+1)\, x^2 y + (x+1)\frac{y^3}{3}\right]_0^x dx$$

$$= \int_0^1 \left[(x+1)x^3 + (x+1)\frac{x^3}{3}\right]_0^x dx = \frac{4}{3}\int_0^1 (x^4 + x^3)\, dx$$

$$= \frac{4}{3}\left[\frac{x^5}{5} + \frac{x^4}{4}\right]_0^1 = \frac{4}{3}\left[\frac{1}{5} + \frac{1}{4}\right] = \frac{3}{5}.$$

10. $\iiint_S yz\, dx\, dy\, dz$: S in the first octant bounded above by $z = 1$ and below by $z = \sqrt{x^2 + y^2}$.

Sol. Since S is in the first octant, the region D in the xy-plane is also in the first quadrant. The surfaces $z = 1$ and $z = \sqrt{x^2 + y^2}$ intersect in the curve $1 = \sqrt{x^2 + y^2}$ in the xy-plane.

$$I = \int_0^1 \int_0^{\sqrt{1-y^2}} \int_{\sqrt{x^2+y^2}}^{1} yz\, dz\, dx\, dy$$

$$= \int_0^1 \int_0^{\sqrt{1-y^2}} \left[y\frac{z^2}{2}\right]_{\sqrt{x^2+y^2}}^{1} dx\, dy$$

$$= \int_0^1 \int_0^{\sqrt{1-y^2}} \frac{y}{2}(1 - x^2 - y^2)\, dx\, dy$$

$$= \int_0^1 \frac{y}{2}\left[x - \frac{x^3}{3} - y^2 x\right]_0^{\sqrt{1-y^2}} dy$$

$$= \int_0^1 \frac{y}{2}\left[\sqrt{1-y^2}\,(1-y^2) - \frac{(1-y^2)^{3/2}}{3}\right] dy$$

$$= \frac{1}{3}\int_0^1 y\,(1-y^2)^{3/2}\,dy = \frac{1}{3}\left[\frac{(1-y^2)^{3/2}}{-5}\right]_0^1 \frac{1}{3}\cdot\frac{1}{5} = \frac{1}{15}.$$

Find the volume of the given solid (Problems 11 – 13):

11. Bounded by the coordinate planes and $\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} + \sqrt{\frac{z}{c}} = 1.$

Sol. Required volume

$$= \int_0^{a} \int_0^{b(1-\sqrt{x/a})^2} \int_0^{c(1-\sqrt{x/a}-\sqrt{x/b})^2} dz\,dy\,dx$$

$$= \int_0^{a} \int_0^{b(1-\sqrt{x/a})^2} c\left(1-\sqrt{\frac{x}{a}}-\sqrt{\frac{y}{b}}\right)^2 dy\,dx$$

$$= \int_0^{a} \int_0^{b(1-\sqrt{x/a})^2} c\left[\left(1-\sqrt{\frac{x}{a}}\right)^2 - 2\left(1-\sqrt{\frac{x}{a}}\right)\sqrt{\frac{y}{b}} + \frac{y}{b}\right] dy\,dx$$

$$= \int_0^{a} c\left[\left(1-\sqrt{\frac{x}{a}}\right)^2 y - 2\left(1-\sqrt{\frac{x}{a}}\right)\frac{2}{3}\cdot\frac{y^{3/2}}{\sqrt{b}} + \frac{y^2}{2b}\right]_0^{b(1-\sqrt{x/a})^2} dx$$

$$= c\int_0^{a} \left[b\left(1-\sqrt{\frac{x}{a}}\right)^4 - \frac{4b^{3/2}}{3\sqrt{b}}\left(1-\sqrt{\frac{x}{a}}\right)^4 + \frac{b^2}{2b}\left(1-\sqrt{\frac{x}{a}}\right)^4\right] dx$$

$$= c\int_0^{a} \left(1-\sqrt{\frac{x}{a}}\right)^4 \left(b - \frac{4}{3}b + \frac{b}{2}\right) dx$$

$$= \frac{1}{6}bc\int_0^{a} \left[1 - 4\sqrt{\frac{x}{a}} + 6\frac{x}{a} - 4\left(\frac{x}{a}\right)^{3/2} + \left(\frac{x}{a}\right)^2\right] dx$$

$$= \frac{1}{6}bc\left[x - \frac{4}{\sqrt{a}}\frac{x^{3/2}}{3/2} + 6\frac{x^2}{2a} - \frac{4}{a^{3/2}}\cdot\frac{x^{5/2}}{5/2} + \frac{x^3}{3a^2}\right]_0^a$$

$$= \frac{1}{6}bc\left[a - \frac{8}{3}a + 3a - \frac{8}{5}a + \frac{1}{3}a\right] = \frac{1}{6}abc\left[4 - \frac{8}{3} - \frac{8}{5} + \frac{8}{5} + \frac{1}{3}\right] = \frac{abc}{90}$$

2. Bounded above by $z = 4 - x^2 - y^2$ and below by $z = 4 - 2x$.

ol. The region D in the xy-plane is the curve of intersection of the surfaces $z = 4 - x^2 - y^2$ and $z = 4 - 2x$

i.e., $4 - 2x = 4 - x^2 - y^2$ or $x^2 + y^2 - 2x = 0$.

Thus $-\sqrt{2x - x^2} \le y \le \sqrt{2x - x^2}$; $0 \le x \le 2$.

Required volume

$$V = \int_0^2 \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} \int_{4-2x}^{4-x^2-y^2} dz\, dy\, dx$$

$$= \int_0^2 \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} (4 - x^2 - y^2 - 4 + 2x)\, dy\, dx$$

$$= \int_0^2 \left[(2x - x^2)\, y - \frac{y^3}{3} \right]_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} dx$$

$$= \int_0^2 \left[2(2x - x^2)^{3/2} - \frac{2}{3}(2x - x^2)^{3/2} \right] dx = \frac{4}{3} \int_0^2 (2x - x^2)^{3/2}\, dx$$

Put $x - 1 = X$ so that $dx = dX$

$$V = \frac{4}{3} \int_{-1}^{1} (1 - X^2)^{3/2}\, dX$$

Now put $X = \sin\theta$; $dX = \cos\theta\, d\theta$ so that

$$V = \frac{4}{3} \int_0^{\pi/2} \cos^4\theta\, d\theta = \frac{8}{3} \int_0^{\pi/2} \cos^4\theta\, d\theta = \frac{8}{3} \cdot \frac{3}{4\,.\,2} \cdot \frac{\pi}{2} = \frac{\pi}{2}.$$

13. Bounded by the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Sol. Required volume

$$= 8 \int_0^a \int_0^{b\sqrt{1 - x^2/a^2}} \int_0^{c\sqrt{1 - x^2/a^2 - y^2/b^2}} dz\, dy\, dx$$

$$= 8c \int_0^a \int_0^{b\sqrt{1 - x^2/a^2}} \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}\, dy\, dx$$

$$= 8c \int_0^a \int_0^{b/a\sqrt{a^2-x^2}} \left[\frac{b^2}{a^2}\left(\frac{a^2-x^2}{b^2}\right) - \frac{y^2}{b^2}\right]^{1/2} dy\, dx$$

$$= 8c \int_0^a \int_0^{b/a\sqrt{a^2-x^2}} \frac{1}{b}\sqrt{\frac{b^2}{a^2}(a^2-x^2) - y^2}\, dy\, dx$$

$$= \frac{8c}{b}\int_0^a \left[y\sqrt{\frac{b^2}{a^2}(a^2-x^2) - y^2} + \frac{b^2(a^2-x^2)}{2a^2}\arcsin\left(\frac{y}{\frac{b}{a}\sqrt{a^2-x^2}}\right)\right]_0^{\frac{b}{a}\sqrt{a^2-x^2}} dx$$

$$= \frac{8c}{b}\int_0^a \left(0 + \frac{b^2(a^2-x^2)}{2a^2}\cdot\frac{\pi}{2}\right) dx = \frac{2\pi b\, c}{a^2}\left[a^2x - \frac{x^3}{3}\right]_0^a = \frac{4\pi abc}{3}.$$